

1. SQL Basics – Interview Questions

Theoretical Questions

1. What is SQL?

Answer:

SQL (Structured Query Language) is used to communicate with databases to store, retrieve, update, and delete data.

2. What are types of SQL commands?

Answer:

- DDL (Data Definition Language): CREATE, ALTER, DROP, TRUNCATE
- DML (Data Manipulation Language): INSERT, UPDATE, DELETE
- DCL (Data Control Language): GRANT, REVOKE
- TCL (Transaction Control Language): COMMIT, ROLLBACK, SAVEPOINT

3. Difference between DBMS and RDBMS?

Feature	DBMS	RDBMS
Structure	Non-relational	Relational (tables)
Relationships	Not supported	Supported
Example	File system	MySQL, Oracle

4. What is a table?

A collection of data organized in rows and columns.

5. What is a row and column?

- Row: A single record
- Column: Attribute/field of data

⑥ What are data types?

Define the type of data a column can hold.

Examples:

- INT → numbers
- VARCHAR → text
- DATE → date
- FLOAT → decimal numbers

⑦ Difference between DELETE, TRUNCATE, DROP?

Command	Description
DELETE	Removes specific rows
TRUNCATE	Removes all rows (fast)
DROP	Deletes entire table

⑧ What is PRIMARY KEY?

A column that uniquely identifies each record.

⑨ What is NULL?

Represents missing or unknown data.

⑩ What is DEFAULT constraint?

Provides a default value if none is specified.

⑪ What is a constraint?

Rules applied on columns to maintain data accuracy and integrity.

12. Types of constraints?

- NOT NULL
 - UNIQUE
 - PRIMARY KEY
 - FOREIGN KEY
 - CHECK
 - DEFAULT
-

13. What is normalization?

Process of organizing data to reduce redundancy.

14. Types of normalization?

- 1NF → Atomic values
- 2NF → Remove partial dependency
- 3NF → Remove transitive dependency

Commands & Behavior

15. What is the difference between DROP and TRUNCATE?

- DROP → removes table completely
 - TRUNCATE → removes all data but keeps structure
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16. Is TRUNCATE faster than DELETE? Why?

Yes, because it does not log individual row deletions.

17. Can we rollback TRUNCATE?

No (in most DBMS), because it is auto-commit.

18. What is auto-commit?

Every SQL statement is automatically committed without manual control.

19. Difference between DELETE and TRUNCATE?

DELETE	TRUNCATE
Row-wise deletion	Removes all rows
Can use WHERE	No WHERE
Slower	Faster

20. What is a schema?

Logical structure of a database (tables, views, etc.).

21. What is metadata?

Data about data (e.g., table structure).

22. What is a view?

A virtual table created from a query.

23. Difference between table and view?

Table	View
Stores data	Virtual
Physical storage	No storage

24. What is indexing?

Technique to improve query performance.

Data Types Deep Understanding

25. Difference between INT and FLOAT?

- INT → whole numbers
 - FLOAT → decimal numbers
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26. What is VARCHAR?

Variable-length string data type.

27. Difference between CHAR and VARCHAR?

CHAR	VARCHAR
Fixed size	Variable size
Faster	Flexible

28. What is DATE data type?

Stores date values (YYYY-MM-DD).

29. What is NULL vs 0?

- NULL → no value
 - 0 → actual numeric value
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Advanced Basic Concepts

30. What is a composite key?

Combination of multiple columns as a primary key.

31. Can NULL be part of PRIMARY KEY?

No.

32. Can NULL be part of UNIQUE key?

Yes (depends on DBMS).

33. What is candidate key?

All possible keys that can uniquely identify rows.

34. What is alternate key?

Candidate keys not chosen as primary key.

35. What is surrogate key?

Artificial key (like auto-increment ID).

36. What is natural key?

Real-world data used as a key (e.g., email).

37. What is denormalization?

Combining tables to improve performance.

38. What is redundancy?

Duplicate data in a database.

39. What is data integrity?

Accuracy and consistency of data.

Transactions & Control

40. What is a transaction?

A group of SQL operations executed together.

41. What are ACID properties?

- A → Atomicity
 - C → Consistency
 - I → Isolation
 - D → Durability
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42. What is COMMIT?

Saves changes permanently.

43. What is ROLLBACK?

Undo changes.

44. What is SAVEPOINT?

Temporary point to rollback partially.

Interview Tricky Questions

45. What happens if you insert duplicate PRIMARY KEY?

Error occurs.

46. Can a table exist without a primary key?

Yes, but not recommended.

47. Can we have multiple UNIQUE constraints?

Yes.

48. What is default value behavior?

If no value is provided, default is inserted.

49. What is difference between SQL and MySQL?

- SQL → language
 - MySQL → database software
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50. What is case sensitivity in SQL?

Depends on DBMS (MySQL is case-insensitive for keywords).

51. What is the use of ALTER?

Modify table structure.

52. What is data definition?

Structure of database objects.

53. What is data manipulation?

Operations on data (insert/update/delete).

54. Why SQL is important for Data Analysts?

Because it helps to:

- Extract data,
 - Analyze datasets
 - Generate reports
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Practical Questions (With Queries)

1. Create a table

Question: Create a table employees

```
CREATE TABLE employees (  
    id INT PRIMARY KEY,  
    name VARCHAR(50),  
    salary FLOAT,  
    joining_date DATE  
);
```

2. Insert data

```
INSERT INTO employees (id, name, salary, joining_date)  
VALUES (1, 'Adarsh', 50000, '2024-01-10');
```

3. Insert multiple rows

```
INSERT INTO employees VALUES  
(2, 'Rahul', 60000, '2023--05-20'),  
(3, 'Amit', 55000, '2022-03-15');  
);
```

4. Select data

```
SELECT * FROM employees;
```

5. Add a new column

```
ALTER TABLE employees  
ADD department VARCHAR(30);
```

6. Modify column

```
ALTER TABLE employees  
MODIFY salary DOUBLE;
```

7. Rename column

```
ALTER TABLE employees  
RENAME COLUMN name TO emp_name;
```

8. Delete specific rows

```
DELETE FROM employees  
WHERE id = 2;
```

9. Remove all data

```
TRUNCATE TABLE employees;
```

10. Drop table

```
DROP TABLE employees;
```

Important Interview Scenarios

1. Create table with constraints

```
CREATE TABLE students (  
    id INT PRIMARY KEY,  
    name VARCHAR(50) NOT NULL,  
    age INT CHECK (age >= 18)  
);
```

```
city VARCHAR(30) DEFAULT 'Delhi'
),
```

2. Insert with NULL

```
INSERT INTO employees (id, name, salary)
VALUES (4, 'Karan', NULL);
```

3. What happens if PRIMARY KEY is duplicated?

It will throw an error (violates uniqueness).

4. Can a table have multiple primary keys?

No, but it can have a composite primary key.

5. Difference between VARCHAR and CHAR?

- VARCHAR → variable length
- CHAR → fixed length

Theoretical Scenario-Based SQL Questions

1. Data Deletion Strategy

Scenario:

You need to remove all data from a table quickly, and you don't need rollback.

Question:

Which command will you use and why?

Answer: Use TRUNCATE because it is faster and removes all rows without logging individual deletions.

2. Data Safety Scenario

Scenario:

You accidentally deleted some records and want to recover them.

Question: Which command could have helped?

Answer: ROLLBACK (if transaction was not committed).

3. Duplicate Data Issue

Scenario:

Your table has duplicate records.

Question: How will you prevent duplicates in future?

Answer: Apply UNIQUE constraint or PRIMARY KEY.

4. Missing Data Problem

Scenario:

Some columns have missing values.

Question: How do you handle it at database level?

- Use DEFAULT values
 - Use NOT NULL constraint
 - Handle using functions like COALESCE
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5. Data Integrity Scenario

Scenario:

You want to ensure that every employee must belong to a valid department.

Question: What will you use?

Answer:

Foreign Key constraint to maintain referential integrity.

6. Performance Issue

Scenario:

Your query is slow when searching by salary.

Question: What can improve performance?

Answer: Create an INDEX on the salary column.

7. Table Design Scenario

Scenario:

You are designing a student table.

Question: How will you ensure each student is uniquely identified?

Answer: Use a PRIMARY KEY.

8. Multiple Column Uniqueness

Scenario:

A combination of two columns must be unique.

Use COMPOSITE PRIMARY KEY or UNIQUE (col1, col2)

9. Data Consistency Scenario

Scenario:

Multiple users are updating the same data.

Question: How to ensure consistency?

Answer:

Use transactions + ACID properties

10. Rollback Partial Changes

Scenario:

You want to undo part of a transaction.

Question:

Use transactions + ACID properties

Intermediate Theoretical Scenarios

Scenario:

You need to store phone numbers.

Question:

Use VARCHAR (not INT) because:

- May contain leading zeros
- Not used for calculations

12. Fixed vs Variable Data

Scenario:

You store country codes like "IN", "US".

Answer:

Use CHAR(2) (fixed length)

13. Table Removal Decision

Scenario:

You want to completely remove table structure.

Answer:

Use DROP

14. Partial Data Removal

Scenario:

You want to delete only some records.

Answer:

Use DELETE WITH WHERE

15. Default Value Scenario

Scenario:

If no city is provided, default should be "Delhi"

Answer:

Use DEFAULT 'Delhi'

Advanced Theoretical Scenarios

16. Normalization Need

Scenario:

Your table has repeated data.

Question: What should you do?

Answer:

Apply **Normalization** (combine redundancy).

17. Denormalization Case

Scenario:

You need faster reporting queries.

Answer:

Use **Denormalization** (combine tables)

18. Null Handling Decision

Scenario:

Email is unique but may change.

Question: Should it be primary key?

Answer:

No → use surrogate key (ID)

20. Data Validation

Scenario:

Age must be greater than 18.

Answer:

Use CHECK (age > 18)

Tricky Interview Scenarios

21. Can a table have no primary key?

Answer: Yes, but not recommended.

22. Can NULL be compared using = ?

Answer: No → use IS NULL

23. Why is TRUNCATE faster than DELETE?

Answer: Because it doesn't log row-by-row deletion.

24. Why is TRUNCATE faster than DELETE?

Answer: No

25. Can we use WHERE in TRUNCATE?

Answer: No

25. What happens if foreign key is violated?

Answer: Database will throw an error.

Query Scenario-Based SQL Questions

1. Employee Salary Update Scenario

Question:

You need to increase salary by 10% for employees in the IT department.

```
UPDATE employees  
SET salary = salary * 1.10  
WHERE department = 'IT';
```

Tests: UPDATE + WHERE

2. Missing Data Scenario

Question:

Find employees whose salary is not provided.

```
SELECT *  
FROM employees  
WHERE salary IS NULL;
```

Tests: NULL handling

3. Duplicate Data Scenario

Question:

Find duplicate emails in a users table.

```
SELECT email, COUNT(*)  
FROM users  
GROUP BY email  
HAVING COUNT(*) > 1;
```

Tests: GROUP BY + HAVING

4. Data Cleaning Scenario

Question:

Replace NULL salary with 0.

```
SELECT name, COALESCE(salary, 0) AS salary  
FROM employees;
```

Tests: COALESCE

5. Table Modification Scenario

Question:

Add a column bonus to employees.

```
ALTER TABLE employees  
ADD bonus INT;
```

Tests: ALTER

6. Business Insight Scenario

Question:

Find total salary expense per department.

```
SELECT department, SUM(salary)  
FROM employees  
GROUP BY department;
```

Tests: Aggregation

7. Filtering Groups Scenario

Question:

Find departments with more than 5 employees.

```
SELECT department, COUNT(*)  
FROM employees  
GROUP BY department  
HAVING COUNT(*) > 5;
```

Tests: Aggregation

Tests: HAVING

8. Relationship Scenario

Get employee names along with their manager names.

```
SELECT e.name AS employee, m.name AS manager
FROM employees e
LEFT JOIN employees m
ON e.manager_id = m.id;
```

Tests: Self JOIN

9. Data Deletion Scenario

Delete employees who left the company (status = 'inactive').

```
DELETE FROM employees
WHERE status = 'inactive';
```

Tests: DELETE

10. Table Reset Scenario

Remove all data but keep table structure.

```
TRUNCATE TABLE employees;
```

Tests: TRUNCATE

11. Top Performer Scenario

Find highest paid employee.